

COMP 130 Lab Assignment #9

Numerical Integration

Due at the start of your next lab

1 Objectives

- Implement the trapezoidal rule and Simpson's rule for numerical integration as MATLAB functions.
- Implement a general-purpose solver, i.e., a function that gets all information about the specific problem as arguments to the function.
- Implement a script file that uses the above functions and demonstrates the theoretical accuracy of the methods and the effects of finite-precision computation.
- Practice functions: *abs*, *loglog*, *error*, *sum*, *rem* and *fix*

2 Background

In engineering, we may need to integrate functions that are difficult or impossible to integrate in a closed form. In other cases, we might need to integrate experimental data. Numerical integration or quadrature can be used to estimate the integral of a function.

A common integration method is the trapezoidal rule, where the region to be integrated is divided up into a number of small intervals and the area of each interval is estimated using a trapezoid. For a single trapezoid, we have

$$\int_a^b f(x) \approx \frac{b-a}{2} [f(a) + f(b)]$$

If we use N trapezoids (as we do in this lab), the approximation becomes

$$\int_a^b f(x) \approx \frac{b-a}{2N} [f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + f(x_N)]$$

where $x_k = a + kh$ and $h = (b-a)/N$.

With Simpson's Rule, we use a quadratic polynomial to estimate the area for each pair of intervals. If we use two intervals i.e., a single pair, Simpson's approximation is

$$\int_a^b f(x) \approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Of course, the accuracy would improve if we used more than two intervals! If we use N intervals (i.e., $N/2$ pairs of intervals), the total area is given by

$$\int_a^b f(x) \approx \frac{h}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + f(x_N)]$$

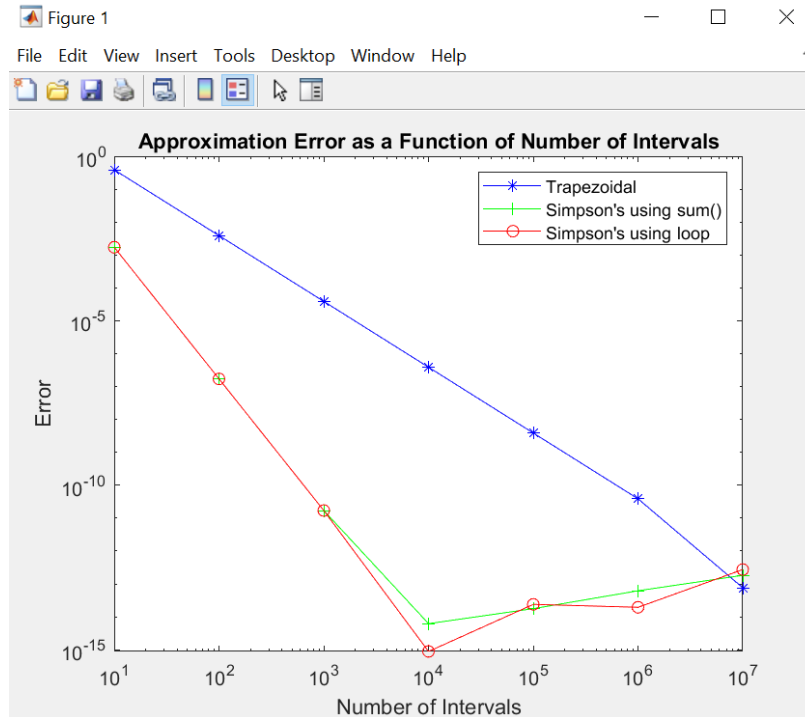
where $x_k = a + kh$, $h = (b - a)/N$, and N must be a positive even number.

3 Requirements

1. The MATLAB code for the trapezoidal rule using double-precision arithmetic (the default) [trapezoidalRule.m](#) and the testing script [integrationError.m](#) have been provided with this lab.
2. Using the example code as a guideline, create a function file named [simpsonsRuleSum.m](#) that implements Simpson's rule. Include the input-data error checking that is used in [trapezoidalRule.m](#) and add checks to ensure that N is even and $N > 1$. (Note that x_0 and x_N in this function are called a and b in the mathematics above.)

The implementation of this function should be based on the MATLAB [sum](#) function without using any [for](#) or [while](#) loops.

3. Create a function file named [simpsonsRuleLoop.m](#) that implements the Simpson's rule. Unlike the previous task, the implementation should be based on loop and if-else without using the MATLAB [sum](#) function.



4. Once the above two function files are created, the testing script [integrationError.m](#) will plot the error of your estimates of the exact integral on a log-log plot with markers that show the location of the individual points (see figure above). You should get fairly straight lines as shown in the figure.

5. Try a different function in the testing script for integration.

4 Evaluation

1. Show the instructor your figure. You may need to use the MATLAB [rem](#) function for this lab.
2. Submit your lab files to D2L → My Tools → Assignments. Hand in your MATLAB file [simpsonsRuleSum.m](#) and [simpsonsRuleLoop.m](#) with comments. You do not need to hand in the plot, just the code.